

23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B.Tech CSE

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

Group No :C15

Team Members :

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1. Problem Title : Smart Parking Management System

2. Problem Description : Urban areas have a rapid increase in vehicle ownership, but parking facilities have not expanded at the same rate. Drivers often spend a lot of time searching for parking spaces in malls, offices, and public areas. Traditional parking systems rely on manual monitoring and cannot provide real-time information about available slots. This results in traffic congestion, fuel wastage, and driver frustration.

3. Identified Entities (Classes) :

Class Name	Description
User	The one who uses the application(admin or driver)
Driver	The user who uses the application to search, reserve parking slots, and make payments.
Admin	The user who manages the entire parking system.
ParkingLot	The entire parking facility.
ParkingSlot	The individual parking facility.
Reservation	Booking made by the user for the slot.
Payment	Billing and payment.

4. Class Attributes and Behaviours

Class Name	Attributes	Methods(behaviours)
1.User	• userID • name • phoneNumber	• login() • logout()
2. Driver	• vehicleNumber	• searchSlot() • reserveSlot() • makePayment() • viewHistory()
3.Admin	• adminID • email • password • parkingPrice	• manageSlots() • updatePricing() • monitorParking() • generateReports()
4.ParkingLot	• lotID • location • totalSlots • availableSlots • slots	• addslot() • checkAvailability() • updateSlotCount() • getAvailableSlots() • displayLot()
5.ParkingSlot	• slotID • slotNumber • floorNumber • status	• getStatus() • reserveSlot() • releaseSlot() • updateStatus() • diaplaySlot()
6. Reservation	• reservationID • reservationTime • expirationTime • status • User • slot	• createReservation() • cancelReservation() • checkStatus() • displayReservation

7.Payment	• paymentID • amount • paymentMethod	• calculateFee() • processPayment() • generateReceipt()
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